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A UNIQUE BIG RED DOT?: A NITROGEN-LOUD ULTRA-MASSIVE GALAXY AT $z\sim 3.8$*

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ABSTRACT

We present spectroscopic confirmation of an ultra-massive ($\log(M_\star/\mathrm{M}_\odot) = 11.2$) nitrogen emitter at $z \sim 3.8$ using both rest-UV and rest-optical observations with Keck/LRIS and Keck/MOSFIRE. The overall SED shape of this object is synergistic to the unique shape of high-redshift Little Red Dots, while being slightly more massive and thus more extended. We find evidence for a massive black hole through measurements of the broad line region in Hβ. We also find significant NIV]1486 emission, uniquely stronger than CIV. This makes this object a unique N-emitter, both through its unusual mass and strength of NIV].

Subject headings: galaxies:high-redshift – galaxies:evolution

1. INTRODUCTION

The launch of *JWST* has introduced a myriad of new astrophysical phenomenon in terms of high-redshift galaxy evolution. Among those is the study of the evolution of the most massive galaxies in the Universe. In particular, Ultra-massive galaxies (UMGs; $\log M_\star > 11$ M$_\odot$) at $z \gtrsim 4$ pose interesting questions on the rapid formation required to assemble these systems. The MAGAZ3NE survey (see Forrest et al. 2020a; **?**) has targeted > 30 such systems in the recent years. [TO DO: expand UMG introduction – rapid assembly timescales, challenge to ΛCDM, mention MAGAZ3NE results]

Recent *JWST* observations discovered a hidden population of compact, red galaxies, termed so-called "little red dots" (LRDs). These galaxies are spatially compact, have a rising red slope at long wavelength filters (i.e. F277W), and are coupled with a significant blue SED toward the rest-UV (**?**). The majority of LRDs being uncovered with *JWST* imaging are high redshift (i.e. $z > 5$, **?**) though there are lower-redshift analogs. LRDs were first discovered as predominately low-mass galaxies (**?**), with evidence for comparatively massive black holes. The black-hole-to-stellar-mass ($M_{\mathrm{BH}}/M_\star$) ratios in this population is elevated above what we see at lower redshifts, ranging from 30% to almost 110%. However, recent observations have began to uncover a handful of more massive LRD candidates (**?**), which in one instance have measured a smaller $M_{\mathrm{BH}}/M_\star$ of $\sim 1\%$ (see). Thus, there is still a lot of work to be done to understand this new population of galaxies, and in particular studying the more massive regime to fully understand the nature of LRDs.

[TO DO: Introduction to Nitrogen Emitters – briefly describe what N-emitters are, why they are rare, what physical conditions produce NIV] emission (high ionization, N/O enrichment, WR stars, AGN, compact starbursts). Mention handful of pre-JWST detections and the growing JWST census. Note that NIV] > CIV is especially unusual and motivates this work.]

[TO DO: Outline of paper – "In this paper we present..., the paper is organized as follows: Section 2 describes the data, Section 3 the analysis..."]

2. DATA

2.1. *Target Selection and Photometric Measurements*

XMM-VID3-1638 (henceforth UMG1638) was part of a candidate sample of ultramassive galaxies (UMGs) at $z > 3.5$, representing a high-redshift extension of the Massive Ancient Galaxies at $Z > 3$ NEar-infrared (MAGAZ3NE) survey (Forrest et al. 2020a,b). We searched for star-forming UMG candidates ($M_\star \gtrsim 10^{11}$ M$_\odot$) at $3.5 < z_{\mathrm{phot}} < 6.5$, an optimal range for rest-frame UV and optical coverage of prominent emission features (e.g., Lyα and Hβ) observable at optical and near-infrared wavelengths accessible from the ground. Candidates were selected from the UltraVISTA–VIDEO catalog (Muzzin et al. 2013), which includes wide-field, multi-band imaging in the XMM–LSS field (Jarvis et al. 2012). To increase the likelihood of detection, we prioritized objects with apparent K_s-band magnitudes of $K_s < 28$ and star-formation rates (SFRs) exceeding $\gtrsim 30$ M$_\odot$ yr^{-1}.

UMG1638 ($\alpha = 2^{\mathrm{h}}24^{\mathrm{m}}50.99^{\mathrm{s}}$ and $\delta = -05°10^{\mathrm{m}}31.74^{\mathrm{s}}$) has an apparent K_s-band magnitude of $22.4^{+0.6}_{-0.4}$. Prior to this work, preliminary photometric SED fits using EAZY (Brammer et al. 2008) estimated a $z_{\mathrm{phot}} = 4.059$, a SFR of 35.5 M$_\odot$ yr^{-1}, and a median stellar mass of $\log(M_\star/\mathrm{M}_\odot) = 11.2$. For UMG1638, the UltraVISTA catalog reports photometric measurements at grz bands from CFHT/Megaprime, $griz$ bands from CTIO/DECam, $GRIZY$ bands from Subaru/HSC, $YJZK_s$ bands from VISTA/VIRCAM, as well as

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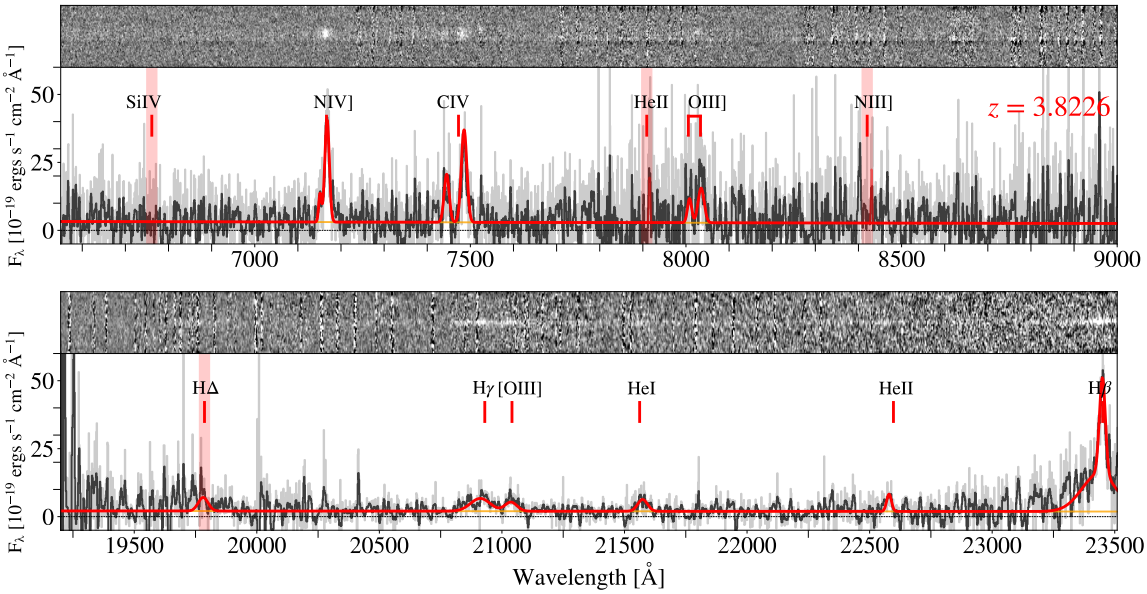


FIG. 1.— (*top*) 1D and 2D spectrum for Keck/LRIS observations. The smoothed spectrum is shown in black and the noise array in grey. The continuum level is shown in orange and the full continuum and emission line model is shown in red. Non-detections and upper limits are shaded in red vertical bars. (*bottom*) Same as above, but for Keck/MOSFIRE observations.

CH1 and CH2 *Spitzer*/IRAC channels, providing a total of 18 photometric bands spanning ~0.4-4.5 μm.

2.2. *Spectroscopic Observations and Reduction*

UMG1638 was first observed in the rest-frame UV with Keck/LRIS-red in October 2021. We employed the 400/8500 grating at a center wavelength of 8000 Å with the 560 dichroic and no dithering between exposures. UMG1638 was placed on a slitmask with slit widths of 1″ and a dispersion of 0.8 Å per pixel. Calibrations for each slitmask on LRIS-red included three internal flat-field frames and an arc lamp spectrum (using Kr, Ar, Ne, and Xe lamps). We observed UMG1638 using individual exposure times of 600 s to mitigate the effect of cosmic rays. The average seeing for these observations ranged from ~0.8–1″ for the first half of the night degrading to > 1″ toward the end of the night with variable transparency. Of the 22 individual frames obtained, the latter half of the night suffered significantly poor conditions, with cloud cover and seeing > 1″ such that the object was undetected in individual frames. We therefore retained 10 science frames with an object detection of S/N > 2, yielding a combined exposure time of 1.6 hours.

The Keck/LRIS observations were reduced using the semi-automatic spectroscopic data reduction code `PypeIt`[1](Prochaska et al. 2020b,a). Each individual exposure was fully reduced (flat-fielded, wavelength calibrated, cosmic ray rejected, and sky-subtracted) producing a 1D and 2D spectra. The 10 reduced frames with S/N> 2 were then coadded in the 2D and the object was extracted using the optimal extraction parameter in `PypeIt` to produce the final 1D and 2D spectrum (see top panel of Figure 1).

UMG1638 was also observed in the *K*-band using Keck/MOSFIRE in September 2022, covering ~19,500–24,000 Å corresponding to rest-frame optical wavelengths (~4050–4970 Å) at z = 3.8226. UMG1638 was observed on a slitmask with a slitwidth of 0.7″ and a dispersion of 2.17 Å per pixel. Standard MOSFIRE calibration frames were taken for flat-fielding and wavelength calibration. We observed UMG1638 for a combined exposure time of 1 hour. The average seeing for these observations was favorable, ranging from ~ 0.45–0.9″.

The Keck/MOSFIRE observations were reduced using the MOSDEF 2D Pipeline[2] (described in Section 3.1 of Kriek et al. 2015). Optimal 1D extraction from the final coadded spectrum was performed using a custom Python code (see Section 2.3.2 in Forrest et al. 2024).

For both spectroscopic observations, we calibrate the flux by a custom Python routine. First the continuum was modeled by masking strong emission lines and smoothing the observed spectrum to trace the underlying continuum shape. We then derive a wavelength-dependent calibration factor to match the observed continuum level to the smoothed continuum of the best-fit modeled SED derived from photometric measurements tracing the stellar continuum only (see more discussion on how this SED was modeled in Section 3.1).

3. ANALYSIS

3.1. *Stellar Mass Measurements*

The overall SED shape has a steep UV slope and large Balmer break. This shape is similar to those of high redshift little red dots, which have been notoriously hard to model in SED fitting methods (e.g. Labbé et al. 2023; **?**; **?**). Coupled with the strong emission lines in the UV and optical, associated with extremely rare environments producing NIV] emission, this object is rare in nature and difficult to reproduce with tra-

[1] https://pypeit.readthedocs.io/en/release/index.html

[2] https://mosdef.astro.berkeley.edu/for-scientists/mosdef-data-reduction-pipeline/

TABLE 1
STELLAR MASS FROM ALTERNATIVE SED FITS USING BAGPIPES, FIT BOTH WITH AND WITHOUT AN AGN COMPONENT.

Star Formation History	without AGN[a]		with AGN[b]	
	$\log(M_\star/\mathrm{M}_\odot)$	A_V [mag]	$\log(M_\star/\mathrm{M}_\odot)$	A_V [mag]
non-parametric	$11.3^{+0.06}_{-0.05}$	$1.03^{+0.12}_{-0.12}$	$11.2^{+0.08}_{-0.07}$	$1.22^{+0.21}_{0.15}$
delayed-τ	$11.22^{+0.09}_{-0.05}$	$0.92^{+0.14}_{-0.21}$	$11.14^{+0.09}_{-0.06}$	$1.07^{+0.16}_{-0.21}$
double-power law	$11.22^{+0.05}_{-0.06}$	$1.06^{+0.11}_{-0.07}$	$11.14^{+0.03}_{-0.04}$	$1.15^{+0.07}_{-0.07}$
Adopted:	Stellar Mass = 11.20 ± 0.06		$A_V = 1.08 \pm 0.10$	

[a] All fits remove the rest-UV bands contaminated by strong emission lines. We reduce the K_s-band flux for fits without an AGN component to isolate the stellar continuum.

[b] Same as above except we retain the flux in the K_s band, as the AGN is expected to contribute significantly to the K_s-band flux (see more detail in Section 3.1).

ditional SED modeling. Thus we use a multifaceted approach to constrain the stellar mass for this object.

UMG1638 exhibits exceptionally strong rest-UV and rest-optical emission lines (Section 3.2; Table 2) that can bias the inferred star-formation history and, in turn, stellar mass (e.g. **?**). Thus, to measure the mass from the underlying stellar continuum alone, we isolate the stellar continuum by identifying and removing/reducing the photometric bands whose transmission curves overlap strong emission features. First we identify photometry that *are not* contaminated by extreme emission lines by overlaying each filter transmission curve on the observed-frame positions of the detected lines. We remove the bands containing any strong rest-UV emission lines (e.g. Lyα, NIV], CIV, OIII). There are only three photometric bands redward of the Balmer break, two IRAC channels and the K_s band. The IRAC channels are placed in such a position to not overlap with any strong emission features. However, the K_s band has coverage of several strong lines (e.g. Hβ, Hγ, [OIII]), which are likely affected by contribution of the AGN. Thus, we use two different approaches to handle this band with fits including and excluding the AGN contribution. For fits without AGN, to anchor the location and strength of the Balmer break (a primary driver of the stellar mass), we keep the K_s photometric point, but reduce the flux in the K_s band to account for the contamination of emission lines in that filter. By applying the K_s transmission curve to the MOSFIRE spectra, we can estimate the integrated flux before and after masking any emission lines at those wavelengths. We find the emission lines are accounting for 26.5% of the flux in the K_s band. We thus reduce the K_s flux by that factor to isolate the stellar continuum. For the fits that *include* an AGN component, we instead retain the full K_s-band flux, as the AGN can contribute substantially to the rest-optical continuum sampled by K_s and reducing this flux would risk remove genuine AGN emission that the AGN component is intended to model. In summary, this leaves a total of 13 photometric bands spanning ~0.4–4.5 μm that represent the underlying stellar continuum of UMG1638.

We then model the SED with a variety of different star-formation histories (SFH), including a non-parametric SFH, a simple τ model, a delayed τ model, and a double-power law. The SED analysis was completed using the python-based SED-fitting code Bayesian Analysis of Galaxies for Physical Inference and Parameter EStimation or BAGPIPES[3] (Carnall et al. 2018). BAGPIPES assumes Bruzual and Charlot stellar populations (Bruzual & Charlot 2003), a Kroupa IMF (Kroupa & Boily 2002), and a Calzetti dust law (Calzetti et al. 2000). In general, the different SFH are fit to 10 – 23 free parameters, allowing for variation in metallicity, age, the slope and strength of the Calzetti dust law, the ionization parameter, and various other parameters. Because the presence of an AGN can substantially alter the inferred stellar mass by contributing flux to the rest-optical continuum (reducing the stellar mass by up to ~2 dex in the most AGN-dominated cases; e.g. **?**), we repeat each SFH fit both with and without an AGN component, using the reduced- and full-K_s photometry respectively as described above.

We find the stellar mass to span $\log(M_\star/\mathrm{M}_\odot) = 11.14 - 11.30$ across all models, largely insensitive to the inclusion of an AGN component (decreasing by only ~0.1 dex on average, despite the AGN fits adopting the full K_s-band flux that maximizes the rest-optical continuum available to the AGN). Stellar mass constraints are summarized in Table 1 and the SED models with the UVISTA photometry are shown in Figure 6. These measurements corroborate that UMG1638 is an extremely massive galaxy, even when conservatively accounting for strong emission lines and AGN activity. We thus adopt for the remainder of the paper the average mass and standard deviation between runs, as $\log(M_\star/\mathrm{M}_\odot) = 11.20 \pm 0.06$.

3.2. *Emission Line Flux and Equivalent Widths*

With both the rest-frame UV and optical observations from Keck/LRIS and Keck/MOSFIRE, we have coverage of a large suite of emission lines, including NIV], CIV, [OIII], and Hβ. We measure the systemic redshift by fitting a single Gaussian profile to the narrow-line component of Hβ, yielding a spectroscopic redshift of $z_{\mathrm{spec}} = 3.8226$. This systemic redshift is then used to anchor the centroids for the remaining emission lines.

To measure the line properties, we first measure the underlying continuum local to the emission line by taking the average of the calibrated flux in clean windows of at least 100 pixels on either side of the line, avoiding skylines. A Gaussian profile is then fit at the observed wavelength of each emission line using the systemic redshift. We take caution with modeling certain emission lines, as described here. The OIII]λ1661,1666 doublet is fit to have consistent line widths between components. The NIV]λ1483,1486 components are fit simultaneously to treat any blending between components with caution. We also model the broad- and narrow-line components for Hβ together and use the velocity width of the broad component to constrain

[3] https://github.com/ACCarnall/bagpipes

the fits to weaker broad Balmer lines (Hγ, fit with a narrow component and Hδ), which are detected at low significance (as they likely trace the same broad-line region). We fit CIV as a doublet as it exhibits broad self-absorption. The auroral [OIII]λ4363 line, which sits on the broad wing of Hγ λ4340, is fit jointly with the broad Hγ component. The fits to each line are shown in Figures **??**.

Line fluxes are computed by integrating the best-fit Gaussian profiles, and rest-frame equivalent widths (EW$_0$) are measured relative to the fitted continuum (emission lines and their associated EW$_0$ and flux measurements are listed in Table 2). Uncertainties on the line fluxes and equivalent widths need to include both the error in our gaussian fits as well how constrained the local continuum is near the line. Therefore we use a Monte Carlo generation of 500 realizations of the spectrum, perturbing the flux at each pixel within its uncertainty and the local continuum level within its uncertainty. The adopted uncertainties correspond to the 16th–84th percentile range of the resulting flux and EW$_0$ distributions.

We caution that, while the line fluxes are robust, the equivalent widths carry a larger systematic uncertainty, due to the continuum being only marginally detected in both spectra (with a S/N ranging ∼1.5-5.8 across the different emission lines). We therefore base our quantitative analysis primarily on the line fluxes and flux ratios, which are insensitive to the continuum placement. We treat any equivalent widths with appropriate caution: for line where the continuum is marginally detected (S/N < 3), but have a clear line detection, we place a lower limit on EW$_0$ and otherwise do not report a equivalent width.

For lines with low S/N (e.g. HeIIλ1640, NII]λ1750, Hδ), we report 3σ upper limits on the flux and EW$_0$. In addition, we caution that Hβ is located at the red-most end of the Keck/MOSFIRE spectrum, such that the tail end of the broad-line profile is cut by ∼XX Å. We estimate the missing flux from the best-fit Gaussian, assuming the broad-line profile is symmetric. Lastly, while there is coverage of Lyα, the line falls at the blue-most end of the LRIS-red coverage, such that it is severely affected by the drop in throughput on the detector, and is therefore not included in our analysis.

3.3. *Star Formation Rates*

As discussed with our stellar mass analysis (Section 3.1), we use a careful, multifaceted approach to measure star-formation rates in UMG1638. Here, we use the measured SFR from different SED models which include all photometry (including those contaminated by emission lines) and the rest-optical spectrum as well as directly measuring SFR from spectral line fitting of Hβ emission.

We first measure the SFR indirectly via SED modeling using `BAGPIPES`. Here, we include the photometry that would be contaminated by emission lines.

To measure the star-formation rate from the Hβ emission, we first separately model the broad-line and narrow-line components. To isolate the star formation derived from stars only (as opposed to AGN contribution), we conservatively use the narrow-line flux. We then measure the SFR directly using:

$$\mathrm{SFR}_{\mathrm{H}\beta} = 1.33 \times 10^{-41} L_{\mathrm{H}\beta} \quad (1)$$

where $L_{\mathrm{H}\beta}$ is in units of erg s^{-1} and SFRs are in units of $M_\odot$ yr^{-1} (Forrest et al. 2020b; Kennicutt 1998; Moustakas et al. 2006). This yields a conservative limit on the SFR, at XX $M_\odot$ yr^{-1}, consistent with a star-forming galaxy.

TABLE 2
DETECTED EMISSION LINES AND THEIR PROPERTIES.

Line	Flux[a]	EW$_0$[b] [Å]	Instrument
NIV] $\lambda\lambda$1483,1486	$52.5^{+6.3}_{-7.1}$	$32.9^{+12.5}_{-9.1}$	LRIS
CIV $\lambda\lambda$1548,1550[c]	$61.0^{+11.2}_{-10.0}$	$32.0^{+22.0}_{-10.0}$	LRIS
HeII λ1640	<35.1	–	LRIS
OIII] $\lambda\lambda$1661,1666	$39.5^{+9.6}_{-7.1}$	>9.9	LRIS
NII] λ1750	<20.4	–	LRIS
Hδ λ4101	<30.7	–	MOSFIRE
Hγ λ4340	$60.2^{+13.1}_{-14.5}$	$53.1^{+12.3}_{-13.7}$	MOSFIRE
[OIII] λ4363	$18.2^{+4.4}_{-6.3}$	$14.2^{+11.1}_{-5.7}$	MOSFIRE
HeI λ4471	$20.1^{+6.7}_{-8.1}$	$23.7^{+10.0}_{-6.8}$	MOSFIRE
HeII λ4685	$17.1^{+6.2}_{-5.5}$	$14.2^{+11.0}_{-6.0}$	MOSFIRE
Hβ λ4861[d]	$209.1^{+26.3}_{-32.3}$	$110.3^{+39.2}_{-33.7}$	MOSFIRE

[a]Fluxes are in units of [$\times 10^{-18}$ erg s^{-1} cm^{-2}]. 3σ upper limits are placed if the S/N of the line drops below 3σ.
[b]EW$_0$ measurements may be systematically uncertain due to low continuum S/N. We thus report a 3σ lower limit if the continuum level is below 3σ threshold but the line is detected. If both are no-detected the EW$_0$ is unconstrained.
[c]The CIV profile exhibits self-absorption, likely due to the BLR in the AGN. Therefore the flux and EW$_0$ measurements are lower limits to the intrinsic values.
[d]The Hβ profile is truncated at the end of the MOSFIRE K-band coverage, we therefore assume symmetry to measure the flux fo the full broad-line component. In addition, the underlying continuum flux is poorly constrained using only the blue end of the line profile,

3.4. *Size and Morphology*

Another key property of LRDs is that they are extremely compact, with sizes roughly XX when measured in the XX band. However, UMG1638 was not observed with *JWST* bands and therefore a direct comparison of sizes measured in XX is not possible. Instead, to measure the size of UMG1638, we use `GALFIT` on the K_s-band. We employ a Sersic profile fit to the data (see Figure 2). We find that the half-light radius is r_e = 3.43 kpc and a Sersic index of n = 0.49. However, the FWHM of the VISTA telescope is 0.51″, so the size results are within the PSF constraints of the telescope. Therefore, the source is point-like in this band, and the size may be taken as an upper limit.

Our measurements determine that UMG1638 is larger than that of a typical LRD. However, given it has a significantly high mass of XX, XX dex more than the typical LRD population, UMG1638 is still extremely compact. Perhaps we are still seeing a system which is LRD-like, but more extended and at lower redshift such that we can begin to differentiate the nature of LRDs. Sources like UMG1638 may thus be helpful in future studies looking at what is the cause of the unique shape of LRDs.

4. AN EXTREME NITROGEN EMITTER

An unusual property of UMG1638 is that it exhibits an extremely-strong NIV] emission in the UV. This type of emission is extremely rare, and has only recently gained momentum in the literature with early galaxy studies from *JWST*. Before *JWST*, only XX nitrogen emitters were known at moderate to high redshifts. *JWST* has discovered many more, leading to

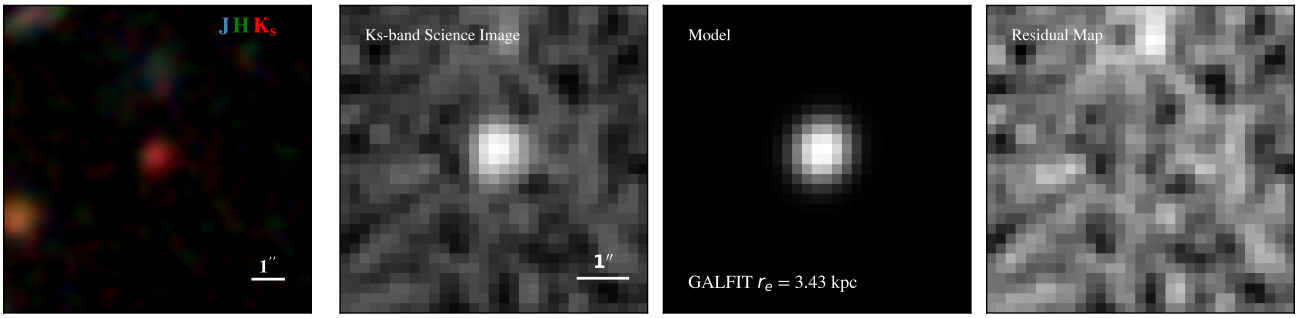


FIG. 2.— GALFIT fitting from UMG1638. (*left*) Science image (K_s-band) used for the profile fitting. (*middle*) Resulting model using a Sersic profile. The half-light radius is r_e = 3.43 kpc and a Sersic index of n = 0.49. The FWHM of the VISTA telescope is 0.51″, so the size results are within the sampling of the telescope. (*right*) The residual map from GALFIT.

XX to date N-emitters at high redshift – still relatively rare, and currently not understood property of high-z galaxies. This makes UMG1638 again a powerful test case to observe NIV] emission at close redshift, within an extremely massive system.

4.1. *NIV] Properties and Line Ratios*

[TO DO: Write this subsection. Describe the NIV]1483,1486 doublet detection (Figure 2 left panel). Report the doublet ratio $\lambda1483/\lambda1486$ = 0.15 and what this implies for electron density (Figure 2 right panel – place on the ne vs ratio diagnostic curves at different Te). Report the derived ne and Te range. Note that NIV]/CIV > 1 is the defining unusual feature here and compare to the handful of other known N-emitters where this ratio has been measured.]

4.2. *AGN vs. Star-Formation Ionization Diagnostics*

[TO DO: Write this subsection. Describe Figure 3 (the OIII]/HeII vs CIV/HeII UV diagnostic diagram from Feltre+2016 and Gutkin+2016). Explain where UMG1638 falls – appears to be in or near the SF region. Discuss what this means: the UV line ratios are more consistent with photoionization by star formation than AGN, despite the broad Hbeta detection. Note the tension and that both may contribute.]

[TO DO: Add the new NIV]-based AGN vs SF diagnostic plot here as a second panel or separate figure. Describe what diagnostic this uses and where UMG1638 falls relative to other N-emitters.]

4.3. *Electron Temperature and Density*

[TO DO: Write this subsection. The NIV] doublet ratio already gives ne (Section 4.1 / Figure 2 right panel). Additionally, [OIII]4363 is detected – discuss what temperature diagnostic this enables. Note that without [OIII]4959,5007 (outside the MOSFIRE K-band window at z=3.82), a direct Te measurement from the standard [OIII] ratio is not possible. However, [OIII]4363/Hbeta can be used as a relative constraint. Report what range of Te and ne are consistent with the data and discuss implications for metallicity (high Te implies low metallicity).]

4.4. *Origin of Nitrogen Emission*

There are several competing explanations for the origin of strong nitrogen emission lines in galaxies. One leading interpretation is that the extreme nitrogen features arise during a highly bursty phase of star formation, occurring when the galaxy is still metal-poor, compact, and dominated by massive stars. Such conditions are met with high-redshift galaxies in the early Universe and are consistent with the growing population of nitrogen emitters recently identified with *JWST*.

A recent study of nitrogen emitters used the specific star-formation rate (sSFR) as a diagnostic of star-formation burstiness. To test this scenario, we compiled all known nitrogen emitters in the literature over the redshift interval $XX < z < XX$. We find that the most massive nitrogen emitters exhibit relatively modest sSFRs of XX, significantly lower than those of lower-mass nitrogen-emitting galaxies. This likely reflects their large stellar masses: nitrogen emission in these systems may originate from compact subregions undergoing short-lived starbursts rather than from galaxy-wide elevated sSFR. Consequently, sSFR alone is an incomplete tracer of the physical processes producing this unusual nitrogen emission.

Intriguingly, the best fit SEDs, which include the spectral emission lines in the rest-optical, estimate the emission lines are due to an intense, recent burst. We posit that the burst is limited to a region that is sufficiently compact such as to create the unusual nitrogen emission line.

Another explanation for nitrogen emission comes from strong AGN activity. Here, we cannot rule out this scenario, as there is sufficient evidence via the broad-line region for an AGN. However, the UV line ratios suggest that the majority of the UV emission lines are due to star-formation – and not AGN activity (see Figure 4). This is consistent with the majority of N-emitters, with the exception of XX. To test this, we use recent theoretical diagnostics to test whether the ionization is from AGN, HII, or ambiguous or both.

[TO DO: Add Wolf-Rayet star discussion if relevant – WR stars can produce the hard ionizing photons needed for NIV] and are associated with young, massive star-forming regions. Check whether the SFR and age constraints from BAGPIPES are consistent with a WR population.]

5. BLACK HOLE AND AGN PROPERTIES

5.1. *Broad Hβ Decomposition*

[TO DO: Write this subsection. Describe the two-component (narrow + broad) Gaussian decomposition of Hbeta (Figure 6 left panel). Report the FWHM of both components: narrow FWHM ≈ 290 km/s, broad FWHM ≈ 1432 km/s. Note that the broad component extends to the red edge of the MOSFIRE detector and that a symmetric Gaussian correction was applied. Mention that the broad component is unambiguously detected at SNR ~ XX.]

5.2. *Black Hole Mass*

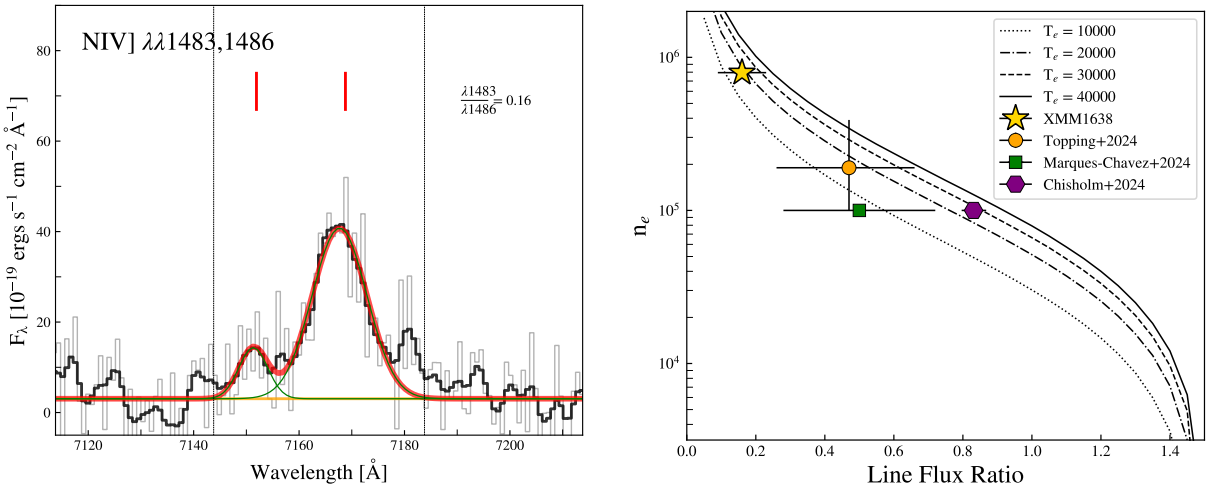


FIG. 3.— (*left*) Zoom-in on the NIV]$\lambda\lambda$1483,1486 doublet. The red line shows the best-fit two-component Gaussian model. The measured doublet ratio λ1483/λ1486 is labeled. (*right*) Electron density diagnostic using the NIV] doublet flux ratio. Curves show the predicted ratio as a function of electron density n_e for electron temperatures T_e = 10,000–40,000 K. The position of UMG1638 is shown as a gold star, alongside other known nitrogen emitters from the literature.

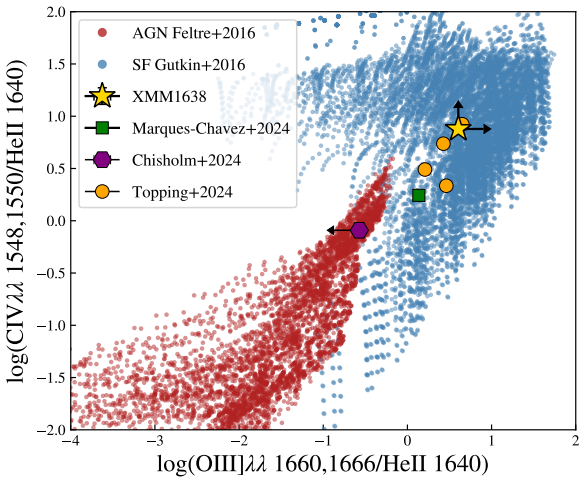


FIG. 4.— UV emission line diagnostic diagram following **?** and **?**, showing log(OIII]$\lambda\lambda$1661,1666/HeIIλ1640) vs. log(CIV$\lambda\lambda$1548,1550/HeIIλ1640). Model grids for AGN (blue) and star-forming (red) ionization are shown. UMG1638 (gold star) is compared to other known nitrogen emitters from the literature.

One interesting property of LRDs found with *JWST* is many seem to harbor massive black holes, such that the black hole to stellar mass ratio sometimes exceeds > 30% (**?**). UMG1638 also has evidence for a black hole, indicated by the broad-line components observed in the MOSFIRE spectrum. To investigate the black hole to stellar mass ratio of UMG1638, we estimate the black hole mass from Hβ emission by using the luminosity and full-width half-maximum (FWHM) of the broad line component of Hβ as in **?**:

$$M_{BH} = (3.6 \pm 0.2) \times 10^6 \left(\frac{L_{H\beta}}{10^{42}\ \mathrm{erg\ s^{-1}}}\right)^{0.56\pm0.02} \left(\frac{FWHM_{H\beta}}{10^3\ \mathrm{km\ s^{-1}}}\right)^2 \mathrm{M_\odot} \quad (2)$$

TABLE 3
SUMMARY OF DERIVED PHYSICAL PROPERTIES OF UMG1638.

Property	Value
R.A. (J2000)	$02^h24^m50.99^s$
Decl. (J2000)	$-05°10^m31.74^s$
z_{spec} [a]	3.8226
K_s [AB mag]	$22.4^{+0.6}_{-0.4}$
$\log(M_\star/\mathrm{M_\odot})$ [b]	11.20 ± 0.06
A_V [mag] [b]	1.08 ± 0.10
$\mathrm{SFR_{H\beta}}$ [$\mathrm{M_\odot\ yr^{-1}}$] [c]	XX
$\log(M_{BH}/\mathrm{M_\odot})$	7.76 ± 0.05
$M_{BH}/M_\star$	~0.05%
$\mathrm{FWHM_{H\beta,\ broad}}$ [$\mathrm{km\ s^{-1}}$]	2107
$\mathrm{FWHM_{H\beta,\ narrow}}$ [$\mathrm{km\ s^{-1}}$]	341
n_e [$\mathrm{cm^{-3}}$] [d]	XX
T_e [K] [d]	XX
r_e [kpc] [e]	< 3.43
Sérsic n [e]	0.49

[a] From narrow Hβ (Section 3.2).

[b] Adopted mean ± std across SED fits (Table 1).

[c] From narrow Hβ flux (Section 3.3).

[d] From the NIV] doublet / [OIII]λ4363 diagnostics (Section 4.3).

[e] From GALFIT on K_s; point-like at VISTA resolution, so r_e is an upper limit (Section 3.4).

The resulting black hole mass for UMG1638 is $\log(M_{BH}/\mathrm{M_\odot}) = 7.76 \pm 0.05$. [TO DO: fix original typo – was written as $M_{BH} = 7.76 \pm 0.05$ $\mathrm{M_\odot}$ which is missing the log. Also fill in the citation for the virial mass estimator.]

5.3. *Black Hole to Stellar Mass Ratio*

The black hole masses from LRDs range from $M_{BH} = 10^{6-8}$ $\mathrm{M_\odot}$, consistent with the black hole from UMG1638. However, UMG1638 does not have an extreme stellar mass to black hole ratio (~ 0.05%, see Figure 5). This can simply be due to the extreme mass of UMG1638, where the typical

stellar mass of LRDs are $M_\star < 10^{10}$ $M_\odot$ (), XX dex lower than UMG1638. This suggests that the black hole within UMG1638 is among the most massive of the LRD population, while the galaxy itself has grown sufficiently in stellar mass to bring the ratio down to values more typical of lower-redshift AGN hosts. UMG1638 is evidence for a galaxy that is similar to an LRD, hosts an extremely massive black hole in its core, but still has sufficiently high normal star formation contributing to its light.

5.4. *AGN Contribution to UV Emission*

[TO DO: Write this subsection. Discuss the AGN power-law continuum contribution from the BAGPIPES fit (f5100, alphalam, betalam posteriors). Quantify what fraction of the UV continuum and line flux may be AGN-driven vs. stellar. Reference the UV diagnostic diagram (Figure 3) showing the lines are more consistent with SF ionization. Discuss the tension: broad Hbeta confirms AGN, but UV lines suggest SF dominates the ionizing photon budget. This may indicate an obscured or low-luminosity AGN whose narrow-line region is dominated by the starburst.]

6. A BIG RED DOT

The overall SED shape of UMG1638 is synergistic to little red dots (LRDs) found in the first years of *JWST*. However, UMG1638 is much more massive and slightly larger than those of the typical population of LRDs. Nonetheless, we note UMG1638 as a "Big Red Dot", a lower redshift, high-mass and more extended analog to LRDs found at earlier times. In the following subsections, we describe the similarities and differences in UMG1638 to the LRD population, and discuss its implications for understanding the origin and properties of high-z LRDs.

6.1. *Photometric Properties*

The SED of UMG1638 has a similar shape as those of the main LRD population, with a rising slope in the rest-UV and significant red flux at > 20000 Å. However, the strictly conventional criteria for LRDs utilize a set of colors and estimated sizes from long-wavelength *JWST* filters. The color-selection criteria for lower-redshift LRDs ($4 < z < 6$) from **?** are as follows:

$$F115W - F150W < 0.8 \quad (3)$$

$$F200W - F277W > 0.7 \quad (4)$$

$$F277W - F444W > 1.0 \quad (5)$$

UMG1638 was not observed with *JWST* filters, so we cannot directly test whether UMG1638 meets these criteria. Therefore, we use the best-fit SED to estimate the flux from different *JWST* bandpasses (here we use the "predict" function in `BAGPIPES`). The resulting measurements are shown in Figure 6 as orange and white crosshairs. These results show that the predicted colors, based on the best-fit SED, satisfy the LRD color criteria within 1σ error, with the exception of criterion (2), in which UMG1638 yields 0.8 mag less than the given threshold. We note that the redshift of UMG1638 is slightly less than that of the redshift range for these criteria ($4 < z < 6$), which may slightly affect the derived colors and may account for this small discrepancy. In summary, the blue UV colors and rising red flux at longer wavelengths are consistent with the LRD population.

6.2. *Comparison to the LRD Population*

[TO DO: Write this subsection. Quantitatively compare UMG1638 to the LRD population in terms of: (1) stellar mass – UMG1638 is XX dex more massive than typical LRDs; (2) size – point-like at VISTA resolution but larger than typical LRDs; (3) MBH/M* ratio – much lower than typical LRDs due to the high stellar mass; (4) redshift – z=3.82 is slightly lower than the main LRD population. Discuss whether UMG1638 represents a more evolved version of an LRD, or a physically distinct population.]

7. WHAT IS UMG1638?

[7.1 What is UMG1638? – Synthesis of all results. A massive, star-forming galaxy at z=3.82 with a Type 1 AGN embedded in a starburst, producing unusual nitrogen emission. Compact enough to be LRD-like but more massive and extended. Possibly caught in the act of quenching or transitioning from starburst to quiescent.]

[7.2 Implications for the N-emitter population – UMG1638 is unusual among N-emitters in its mass. Does nitrogen emission scale with mass or with compactness/sSFR? Place UMG1638 on any sSFR vs. mass diagram of N-emitters from the literature. Discuss whether the compact burst scenario or AGN scenario is more consistent with the overall properties.]

[7.3 Implications for LRD/BRD population – What does UMG1638 tell us about more massive LRDs? Does the lower MBH/M* ratio suggest that stellar mass grows faster than BH mass at high mass? Discuss whether pre-JWST surveys like MAGAZ3NE are finding the high-mass tail of the LRD distribution.]

8. CONCLUSIONS

In this work, we present the spectroscopic confirmation and multi-wavelength characterization of UMG1638, an ultra-massive ($\log(M_\star/M_\odot) = 11.2$) nitrogen-loud galaxy at $z = 3.8226$. Here, we summarize the main results of our analysis:

- [TO DO: Spectroscopic confirmation at z=3.8226 via narrow Hbeta; XX emission lines detected across rest-UV (LRIS) and rest-optical (MOSFIRE).]
- [TO DO: Stellar mass consistent at $\log(M_\star/M_\odot) \sim$ 11.0–11.2 across all SFH models, confirming UMG classification even accounting for AGN and emission line contamination.]
- [TO DO: NIV]λ1486 uniquely stronger than CIV – NIV]/CIV > 1 – placing UMG1638 among the most nitrogen-loud galaxies known. UV diagnostic diagrams favor SF over AGN ionization.]
- [TO DO: Electron density from NIV] doublet ratio: $n_e \sim$ XX cm^{-3} at $T_e \sim$ XX K.]
- [TO DO: Broad Hβ (FWHM ~ 1432 km/s) confirms a Type 1 AGN. Black hole mass $\log(M_{BH}/M_\odot) = 7.76 \pm 0.05$, giving $M_{BH}/M_\star \sim 0.05\%$, well below the LRD population average.]
- [TO DO: SED shape consistent with LRD color criteria, making UMG1638 a "Big Red Dot" – a massive, lower-redshift analog to the LRD population.]

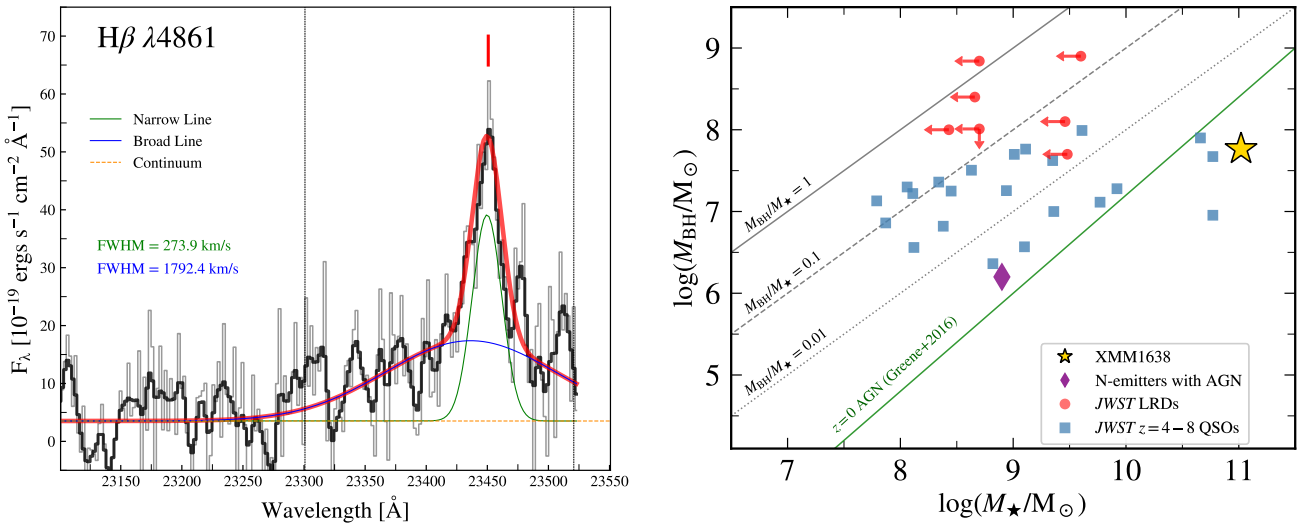


FIG. 5.— (*left*) Two-component decomposition of Hβ into narrow (green) and broad (blue) Gaussian components. The combined model is shown in red and the continuum level as the orange dashed line. The measured FWHM of each component is labeled. (*right*) Black hole mass vs. stellar mass for UMG1638 (gold star) compared to $z = 0$ AGN (**?**), *JWST* LRDs, $z = 4$–8 quasars, and other nitrogen emitters with AGN. Lines of constant $M_{BH}/M_\star$ ratio are shown.

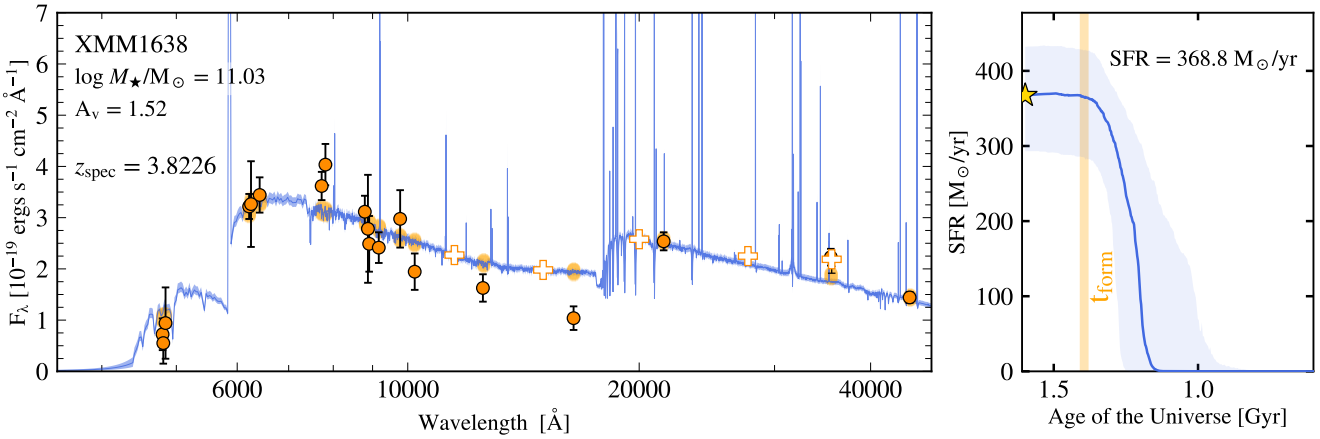


FIG. 6.— [TO DO: Update caption. Left panel: best-fit BAGPIPES SED with photometry (filled orange circles) and predicted JWST photometry (orange/white crosshairs). Right panel: star-formation history posterior.]

ACKNOWLEDGEMENTS

The authors wish to recognize and acknowledge the very significant cultural role and reverence that the summit of Maunakea has always had within the indigenous Hawaiian community. We are most fortunate to have the opportunity to conduct observations from this mountain.

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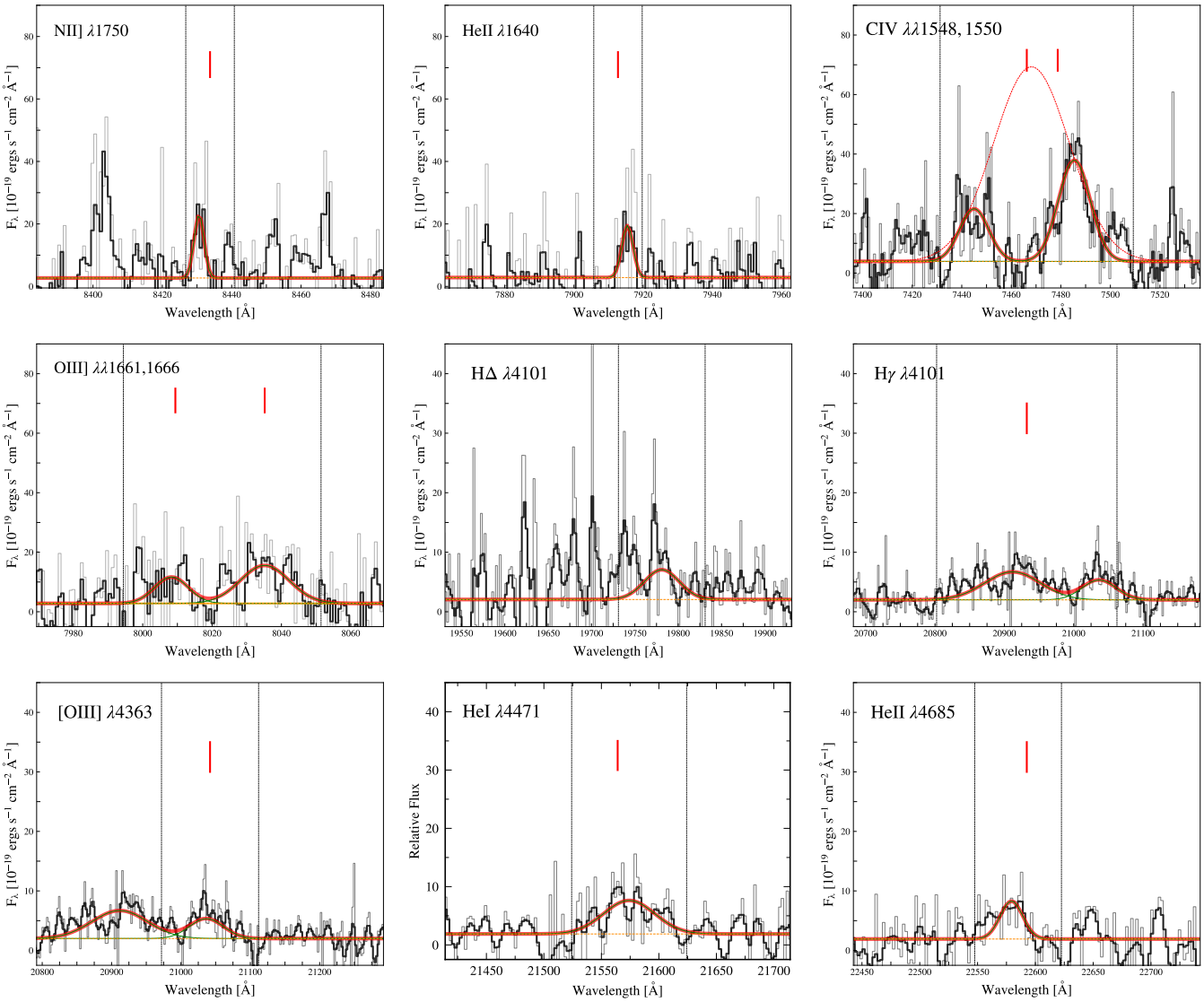


FIG. 7.— Additional emission line fits. Lines and instrument used are labeled at the top left and right (respectively) in each panel. The red line shows the final fit for each line, comprised of either a single or multiple components (each individual component is modeled by a Gaussian fit shown in green). The estimated continuum is shown as the orange dashed line.

APPENDIX

ADDITIONAL EMISSION LINE FITS

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